

UNIT - 01 TECHNIQUES AND RESOURCES OF MATHEMATICS TEACHING

Concept of Mathematics Teaching: Importance of Mathematics in the Field of Education

The importance of mathematics in the field of education is absolute. Through mathematics, the mental, physical, and spiritual development of students takes place. Mathematics is that science of numbers, bases, signs, and brief symbols with the help of which quantities, direction, and space are comprehended.

If all these objectives along with linking it to our day-to-day experiences are integrated into the process of the classroom environment, it will bring a major shift in the mathematical classroom processes. This curriculum attempts to reflect and discuss all of these components. For effective teaching, there must be a deep understanding of mathematical concepts and a clear realization of what the actual nature of mathematics is. NCF 2005 (referred as 2006 in reference context) and BCF 2008 advocate that children are naturally curious, learn continuously, and actively construct knowledge. They know a lot before starting formal schooling, and every child is unique due to their specific social and cultural background, despite having some common qualities.

A teacher understanding the children, recognizing their unique backgrounds, and utilizing their existing knowledge inside the classroom will facilitate and enrich their learning journey.

Objectives of Mathematics Teaching:

1. To understand the methods of learning and teaching mathematics and the various aspects coming under them.
2. To understand the requirements and utility of different types of Teaching Learning Materials (TLM).
3. To understand Unit Planning and Lesson/Learning Planning.
4. To understand concepts related to geometric shapes and patterns.
5. To understand the necessity and application of fractions.
6. To understand operations on fractional numbers.
7. To understand different types of fractions and their expression in decimal numbers.

A Potential Sequence of Learning Mathematics (E-L-P-S)

There is a potential established sequence for learning mathematics, which we refer to as **E - L - P - S** (Hindi acronym: अ - भा - चि - प्र). This stands for:

- **E** - Experience (अनुभव)
- **L** - Language (भाषा)
- **P** - Pictures (चित्र)
- **S** - Symbols (प्रतीक)

We can systematically understand these aspects through the following detailed points:

If we want to teach mathematics to a child, we can facilitate learning through the use of concrete objects. This allows the child to learn easily, quickly, and the learning becomes permanent.

- **Experience (अनुभव):** While teaching mathematics to a child, if we use concrete objects, the child gathers direct experiences. For example: pebbles, stones, or any tangible material.
- **Language (भाषा):** If we try to make the child learn mathematics using their mother tongue, they will learn concepts easily and quickly. This significantly enhances their mathematical skills. For this purpose, we actively use words, contextual stories, and short poems.
- **Pictures (चित्र):** When we teach mathematics and display images or visual representations related to the mathematical concept to the children, their retention and cognitive experience expand greatly. For instance, if we are teaching counting to a child, showing one pen instead of just writing the numeric "1", and showing two pens for "2" increases the child's processing, experience, and knowledge, enabling them to learn effortlessly.
- **Symbols (प्रतीक):** In the sequence of teaching mathematics, we teach a vast array of conceptual ideas through numbers and symbols. For instance, writing a half as $\$1/2\$$ or writing specific digits to denote specific counts (e.g., $* + * = 2$).

Teaching Formal Mathematics by Linking with Concrete Experiences

Formal mathematics refers to that system of mathematics where we perform structured mental calculations or try to resolve complex problems and mathematical questions. However, to resolve these problems effectively, concrete things (real-world objects) are necessary, as they allow us to comprehend those problems with much greater clarity.

For Example: If we want to develop the foundational understanding of numbers in students within a classroom, we can cultivate this numeric understanding using concrete objects and experiences, such as pieces of stones, pens, copies, chocolates, etc.

Furthermore, if a child needs to be taught the concept of a half ($\$1/2\$$), we can explain it using the concrete experiences surrounding them. For instance, we can take out a roti from a child's tiffin box, break it down right through the middle, and show them clearly that this represents a half ($\$1/2\$$).

Teaching Through Play-Way Methods (Playful Learning):

Play is a word that immediately triggers interest and entertainment within every child upon hearing it. Children are perpetually eager to play games, and through this enjoyable medium, they grasp complex concepts which become permanently retained in their minds. Children do not easily forget things learned through playful methods. Often, students develop a sense of fear towards mathematics, but if we teach those exact same concepts through structured games, they take a deep interest in it and their anxiety disappears. Through games, children effortlessly acquire a solid grasp of counting, addition, subtraction, division, etc. Multiple active strategies are implemented via classroom play. For instance, chocolates, pens, or pencils can be used to teach counting. Similarly, an understanding of geometric shapes can also be introduced dynamically through games.

Learning Through Repetition (Iterative Reinforcement):

Children learn dynamically through repetition. Learning via repetition does not mean mindless rote memorization. During childhood (at an early age), children repeat words heard from elders and emulate the behaviors and actions

performed by them. In other words, children learn by imitating their elders. Dropping and picking up an object repeatedly, listening to the same story multiple times, or watching the same visual activity repeatedly are examples of this. Repetition must be designed to be engaging, interesting, and varied so that children experience genuine joy in learning.

Children Learn From One Another (Peer Learning):

Children learn from each other. When children play together, they learn a massive amount from their peers through play. They interact with one another without any inhibitions or fear, and they get an autonomous opportunity to express their thoughts freely. Children pick up concepts rapidly and smoothly through shared stories, rhymes, and interactive games.

Learning From Mistakes (Trial & Error):

Learning from mistakes implies that making errors is an absolute requirement for natural learning. Children's mistakes are a completely natural and organic part of their learning process. Committing mistakes is essential because until children make mistakes, they will not learn. Children's errors give teachers a clear window into how they think and how they process information. Children learn every functional task by making mistakes, rectifying them, and moving forward.

For Example: If we ask a child to write the digit 8, and the child accidentally writes 6 instead of 8, we can gently guide them and show them that this is not 8, clarifying the structure.

Learning Through Activities (Activity-Based Learning):

Through games and play, we conduct multiple structured activities for students. Children learn very rapidly and effortlessly through direct activities. In this framework, they are given ample opportunities for hands-on, experiential learning ("learning by doing"), which naturally captures their interest. This ensures that students are thoroughly active participants. Through activity-based learning, we can clarify foundational concepts such as big-small, inside-outside, above-below, less-more, far-near, and light-heavy by linking them directly to their daily lives.

Mathematical Puzzles: Puzzles hold an independent, significant value in mathematics. Puzzles inspire and motivate students to seek creative solutions. Another positive aspect of puzzles is that while solving them, students have to discover the analytical path to the solution entirely on their own. Solving puzzles actively builds and expands critical thinking faculties in students. It opens up various methodologies for students to look for problem-solving strategies. Many games and puzzles might not be explicitly centered on rigid academic math, but they offer immense joy in execution. For example: arranging the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 inside a grid/square such that the sum of the numbers in every vertical column, horizontal row, and both diagonals equals exactly 15.

Open-Ended Questions and Problems:

When we ask verbal questions inside the classroom that allow for diverse thinking, these are termed open-ended questions. They are also widely known as word problems or situational questions. Such questions encourage and motivate students to provide a variety of creative answers, fostering deep interest. Open-ended questions establish a collaborative classroom environment where teachers receive genuine student feedback, highlighting exactly where instructional adjustments are required. Open-ended questions are defined as questions that possess multiple correct

pathways and answers. Conversely, closed-ended questions are those where students are constrained to reply with a rigid option, such as a simple "yes" or "no".

Diverse Resources for Teaching-Learning Mathematics:

Mathematics generates new creative imaginations within the human brain and provides a logical direction to thoughts. It is inherently connected to abstract elements, signs, symbols, and imaginative concepts. There are countless things in our immediate surroundings that help us learn mathematics effectively. Counting can be learned using chocolates, pens, pencils, and smooth stone pieces. We can learn to draw a perfect circle using a single bangle. We can learn to form distinct angles using tiny wooden sticks, and understand rectangles through notebooks. A cuboid can be easily explained using a matchbox, a cone using an ice-cream cone, and a cylinder using a round cylindrical box or container. Various geometric shapes can be drawn effortlessly on paper. Not all mathematical knowledge can be clarified merely through verbal explanation; rather, abstract concepts must be demonstrated with the vital assistance of concrete objects. Any teaching material or visual aid used to clarify a teaching point or concept is termed **Teaching Learning Material (TLM)**. A good TLM must be user-friendly, cost-effective, and easy to operate.

UNIT - 02 LEARNING PLAN AND ASSESSMENT IN MATHEMATICS

Learning Plan in Mathematics

Children learn optimally only when the curriculum and lessons are arranged and structured systematically and meticulously, ensuring that the process of learning brings joy. Facing well-designed questions assists them in learning and highly motivates them to explore further.

How to Create a Plan:

While drafting a learning plan, it is vital to keep the following core considerations in mind:

- The sequential flow of teaching must align perfectly with the developmental level of the children. This means the entire textbook content should be organized logically.
- For instance, most standard textbooks introduce numbers from 1 to 100 before teaching mathematical operations like addition and subtraction. However, in reality, children build a basic functional understanding of adding and subtracting things before completely memorizing numbers up to 100.
- In fact, addition and subtraction serve as excellent tools to help them compare different groups of shapes and build a strong number sense. Therefore, it is much better to ensure children fully grasp small numbers first before introducing complex arithmetic calculations.

Why Create a Plan?

1. Creating a plan helps in bringing total clarity to educational goals and learning targets.
2. Time can be allocated and distributed efficiently based on contextual classroom requirements.
3. Planning ensures that every crucial and essential aspect of the core subject matter is comprehensively covered.
4. It assists in sequentially arranging the vital learning experiences necessary for true comprehension.
5. It helps in sustaining students' interest, attention, and engagement throughout the lesson.
6. Planning significantly boosts the teacher's self-confidence and brings genuine satisfaction and joy to their work.

Assessment of Mathematics Learning

Assessment refers to those systematic processes that are conducted to measure the learning achievements and progress of a learner. The core objective of assessment is to track children's academic progress and provide targeted guidance to improve their learning outcomes.

Assessment determines the precise extent to which a learner's foundational capacities and skills have developed. Through assessment, specific learning gaps and learning difficulties faced by students are identified, paving the way for proper corrective solutions. In the context of mathematics, when we assess a child inside the classroom, we discover exactly how much the child is understanding and where they are getting confused. Only after obtaining this data can we adjust our teaching style, because unless we identify children's specific difficulties, we cannot resolve their learning problems.

For Example: Suppose we assign an addition problem to a child, and the child commits an error. Conducting a systematic assessment reveals not just that the child made an error, but precisely **why** they made it. The assessment might reveal that the child erred in addition simply because their foundational understanding of counting was incomplete.

Continuous and Comprehensive Evaluation (CCE) in Mathematics at the Primary Level:

Continuous and Comprehensive Evaluation covers both the scholastic (academic) and co-scholastic (co-curricular) aspects of student development. Continuous evaluation refers to assessment that is regular, ongoing, and persistent throughout the instructional cycle. The term 'continuous' implies that evaluation is fully integrated into the entire teaching-learning process. 'Comprehensive' evaluation implies that the assessment covers all developmental domains of the student—including cognitive, affective, and psychomotor domains. CCE is an inseparable part of learning, and both continuous and comprehensive processes run side by side. The core objective of CCE is to enable students to construct meaning, establish a deep understanding of knowledge, and make their learning permanent using their own unique strategic pathways.

Students must be provided with multiple diverse opportunities across various situations to solve mathematical problems and construct new problems on their own. We must introduce students to multiple problem-solving methods so that they develop the capacity to formulate their own unique answers, and also try to analyze and understand the solutions generated by their peers. It is vital that students become completely fearless regarding making mistakes and build the self-confidence to express their mathematical understanding openly. At the primary level, continuous and comprehensive evaluation means ensuring the holistic, all-round development of all scholastic and co-scholastic facets of a child.

Techniques of Assessment: Assessment provides both a quantitative and qualitative description of student behavior. To interpret and analyze student behavior effectively, various tools and assessment instruments are required. Testing is one such tool used to evaluate an individual or a group, providing structured and systematic data. Different types of tests are systematically applied to conduct different forms of assessment.

Nature of Testing / Types of Tests:

1. Oral Test (मौखिक परीक्षण)
2. Written Test (लिखित परीक्षण)
3. Practical/Experimental Test (प्रयोगात्मक परीक्षण)

Understanding Learning Indicators in Mathematics:

Learning indicators show us exactly how much knowledge and skills a child has acquired up to a specific point. Furthermore, indicators explicitly map out the progressive direction of a child's development. They present the direct outlook of the teacher and the school towards assessment, showing how assessment standards are determined based on local needs. Indicators tell us whether the teaching-learning process is moving in the right direction.

For Example: How a child is performing in mathematics and how much they are absorbing is discovered through indicators via assessment. If a child has successfully developed a clear understanding of basic mathematical operations such as addition, subtraction, multiplication, and division, it indicates that their foundational understanding of mathematics has been successfully established.

UNIT - 03 GEOMETRIC SHAPES AND PATTERNS

Open and Closed Shapes

Open Shapes: Those shapes that do not terminate at their starting point are called open shapes. In other words, they originate from one point and end at a completely different point.

Closed Shapes: Those shapes that originate from a single point and terminate exactly at that very same point are called closed shapes. This means their initial starting point and final endpoint are identical.

Regular and Irregular Shapes

Regular Shapes: Shapes where all interior angles and all sides are perfectly equal are called regular shapes. This means the lengths of all sides and the measures of all angles formed within the shape are completely uniform. Such shapes are two-dimensional (2D).

Irregular Shapes: Those shapes where the lengths of the sides and the interior angles formed within the geometric figures are not equal are termed irregular shapes.

Foundational Geometric Concepts: Point, Line, Ray, and Line Segment

Point (बिन्दु): A point is a geometric entity that possesses no length, width, or thickness. A point is exclusively used to denote a specific, precise location or position. Its position is fixed. For example, the tip of a needle, the tip of a pen, the sharp point of a compass, and stars in the sky appear as points; they show location but their explicit length, width, or thickness cannot be measured.

Line (रेखा): A geometric figure formed by an infinite number of points that can be extended endlessly in both directions. In other words, the figure formed by joining any two points together and extending it infinitely is called a line. A line has no width or thickness; it possesses only length. It is represented symbolically with arrowheads on both ends (e.g., $\overleftrightarrow{AB}$).

Ray (किरण): A ray originates from a single fixed starting point and extends infinitely in only one direction. It possesses exactly one initial endpoint. For example, sun rays, light emitting from a torch, etc.

Line Segment (रेखाखण्ड): The minimum distance between two distinct points is called a line segment. It has two fixed endpoints and a definite length. Infinite points reside within it if extended conceptually, but a segment itself is bounded. For example, the edge of a box, the edge of a table/tube, the edge of a book, etc. (Examples: Line segment AB, BC).

Types of Lines and Line Arrangements:

1. Parallel Lines (समानान्तर रेखा)
2. Perpendicular Lines (लम्बवत् रेखा)
3. Intersecting Lines (प्रतिच्छेदी रेखा)
4. Curved Lines (वक्र रेखा)

5. Transversal Lines (तिर्यक रेखा)
6. Concurrent Lines (संगामी रेखा)
7. Coincident Lines (संपाती रेखा)

Angle (कोण): The inclination or divergence between two rays meeting at a common point is called an angle. The two rays forming the angle are termed the arms of the angle. It is symbolically represented using the angle sign ($\angle$). When two rays meet at a common vertex point, an angle is formed.

Two-Dimensional (2D) and Three-Dimensional (3D) Shapes

Two-Dimensional (2D) Shapes: A two-dimensional shape is a flat, planar figure that possesses only two primary dimensions: length and width. The area and perimeter of 2D shapes can be easily calculated. They possess absolutely no thickness or depth and can be measured across two axes. Examples: Rectangles, squares, triangles, circles, etc.

Three-Dimensional (3D) Shapes: Three-dimensional shapes possess three distinct dimensions: length, width, and thickness (or height/depth). Because they occupy physical space, their volume, curved surface area, and total surface area can be calculated. Examples: Cubes, cuboids, cylinders, cones, etc.

Concept of Symmetry and Patterns

Symmetrical Shapes (सममित आकृतियाँ): When a geometric shape can be divided into two perfectly identical and equal halves, it is called a symmetrical shape. Symmetry is that spatial property where if a shape is folded along a line, it appears identical from all directions and matches up perfectly in every single aspect.

Concept of Patterns (पैटर्न की अवधारणा): The concept of a pattern means observing and following a systematic, sequential arrangement to solve a problem or build a sequence (e.g., 2, 4, 6, 8, ...). Any logical, structured method or design adopted to perform an action or arrange elements is termed a pattern.

UNIT - 04 FRACTIONS AND DECIMAL NUMBERS

Fractions

A fraction is a number that represents a part of a whole. This 'whole' can be a single object or a collective group of objects. For example:

- Whole part = \$1\$
- Half part = $\$1/2$
- One-third part = $\$1/3$
- One-fourth part = $\$1/4$

A fraction consists of two structural components:

1. **Numerator (अंश):** The upper part.
2. **Denominator (हर):** The lower part.

Types of Fractions:

1. **Proper Fraction (सम भिन्न):** When the numerator of a fraction is less than its denominator, it is called a proper fraction. Examples: $\$2/4$, $\$3/6$, $\$5/15$.
2. **Improper Fraction (विषम भिन्न):** When the value of the numerator is greater than or equal to the value of its denominator, it is called an improper fraction. Examples: $\$7/5$, $\$8/2$, $\$12/6$, $\$15/4$.
3. **Mixed Fraction (मिश्र भिन्न):** Such a fraction contains two distinct parts: a whole number part and a proper fraction part. It is formed by combining an integer with a proper fraction, meaning an integer is written alongside the numerator and denominator. Examples: $\$5 \frac{1}{4}$, $\$6 \frac{3}{5}$, $\$7 \frac{8}{9}$.
4. **Like Fractions (सजातीय भिन्न):** Fractions that possess completely identical denominators. Examples: $\$2/4$, $\$5/4$, $\$6/4$, $\$8/4$.
5. **Unlike Fractions (विजातीय भिन्न):** Fractions that possess different, unequal denominators. Examples: $\$5/3$, $\$2/8$, $\$1/7$, $\$5/8$.

Application of Fractions in Daily Life:

Fractions hold an immense, specific importance in our daily life. Any calculation, whether small or large, involves fractions. Fractions are used to compute and measure a specific portion of a large group. They are essential for dividing any item equally among people, purchasing items from a shop, and publishing academic examination results. We also use fractional notations to express mathematical division problems cleanly.

Different Meanings of Fractions:

Fractions can be expressed and understood in various mathematical terms. In mathematics, a fraction is not merely viewed as a single type of number; it is known and applied in multiple operational roles:

- Fractions as a form of division (भाग के रूप में)

- Fractions as a ratio (अनुपात के रूप में)
- Fractions as a unit of measurement (मापन के रूप में)
- Fractions as a rational number (परिमेय संख्या के रूप में)

Mathematical Operations on Fractions

1. **Rule of Fractional Addition:** The sum of two fractional numbers results in a fraction.

Example: $\frac{1}{2} + \frac{3}{4} = \frac{7}{4}$

2. **Rule of Fractional Subtraction:** Subtracting two fractional numbers does not always yield a fractional value.

Example: $\frac{3}{2} - \frac{1}{2} = \frac{2}{2} = 1$ (which is an integer, not a fraction).

3. **Rule of Fractional Multiplication:** Multiplying two fractions always results in a fractional number. In this operation, the numerator is multiplied directly by the numerator, and the denominator is multiplied directly by the denominator.

Example: $\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$ or $\frac{4}{2} \times \frac{4}{3} = \frac{16}{6}$

4. **Rule of Fractional Division:** To perform division on fractions, the first fraction is multiplied by the reciprocal of the second fraction (the fraction by which we are dividing).

Example: $\frac{2}{4} \div \frac{3}{5} = \frac{2}{4} \times \frac{5}{3} = \frac{10}{12}$

Decimal Numbers and Decimal Fractions

Decimal Number: Any number that contains a decimal point (.) is called a decimal number. Decimal numbers are extensively used to find precise statistical averages of numbers. The decimal point is a notation symbol to write mathematical numbers. It can be easily rewritten in the form of a fraction. Examples: $\$0.7 = \frac{7}{10}$, $\$8.5 = \frac{85}{10}$.

Decimal Fraction: If the denominator of a fraction is 10 or any power of 10 (like 100, 1000, etc.) and it is written using the decimal notation system, it is called a decimal fraction. "A simple fraction whose denominator is ten or any power of ten is a decimal fraction." For example, when $\frac{7}{10}$ and $\frac{9}{100}$ are written in decimal form, we obtain the decimal fractions 0.7 and 0.09 respectively.

Mathematical Operations on Decimal Numbers:

- **Addition of Decimals:** To add decimal numbers, the decimal points of all numbers are aligned strictly in a straight vertical line, and then the digits are added exactly like ordinary addition. After adding, the decimal point is placed directly down in its aligned position.

Example:

$$\begin{array}{r} 5.43 \\ + 2.35 \\ \hline 7.78 \end{array}$$

- **Subtraction of Decimals:** Subtraction of decimal numbers follows the exact same operations as general subtraction. While writing down the digits, the decimal points must be aligned perfectly in a straight vertical line.

Example:

$$\begin{array}{r} 7.45 \\ - 2.15 \\ \hline 5.30 \end{array}$$

- **Multiplication of Decimals:** First, multiply the numbers as if the decimals are not present. Then, count the total number of digits to the right of the decimal points in both multiplier numbers combined. Finally, count that same number of places from the rightmost side of the product and insert the decimal point.

Example: $\$54.32 \times 1.3 = 70.616\$$ *(Note: Context script mentioned 16.296, corrected mathematically for exact text translation integrity as $\$54.32 \times 1.3 = 70.616\$$)*

- **Division of Decimals:** While dividing decimal numbers, we first convert the numerator and denominator into whole integers by adjusting place values or ignoring the decimal point initially during long division. Then, by counting the decimal places of the dividend, we place the decimal point back into the quotient.

Example: $\$ \frac{25.15}{5} = \frac{2515}{500} = 5.03\$$

Word Problems (इबारती प्रश्न)

Word problems are highly similar to open-ended questions. In this approach, verbal, situational scenarios are presented to children. Often, these questions seem initially complex to students. Word problems should always be framed using clear words and directly linked to their daily life experiences. For instance, asking a child how many total pens or pencils we possess if we merge two groups. To resolve word problems successfully, students must first deeply understand the question to decipher exactly what is being asked and what data is provided.

Significance of Fractions and Decimals in Daily Life:

1. In daily lifestyle transactions, we constantly use fractional terms like a half ($\$ \frac{1}{2}$) or a quarter ($\$ \frac{1}{4}$).
2. Decimals are fundamentally used while adding, subtracting, and accounting for currency, money, and rupees.
3. Fractions are used as key parameters to measure and evaluate the purity of precious ornaments and gold.
4. Fractions and fractional parts are required to understand nutritional metrics and dietary chart planning.
5. While preparing meals for a large gathering, fractions are used to divide and distribute cooking ingredients proportionately among individuals (e.g., adjusting ingredients to serve 10 people).